

Aggregate demand and aggregate supply analysis

Whether the quantity of output an economy changes is determined by whether the total demand for goods and services equals the total supply of goods and services. Demand for an economy's products comes from households, firms and the government within the country and from households, firms and governments in other countries. An economy's output of products is made by producers in the country. Some of these will be in the private sector and some will be in the public sector.

Aggregate demand

The word 'aggregate' means total. Aggregate demand (AD) in economics is used to describe the total spending of consumers (households), firms and the government plus foreigners' spending on the country's exports minus spending by the country's consumers, firms and government on imports.

Aggregate demand consists of four components, as shown in Figure 17.2.

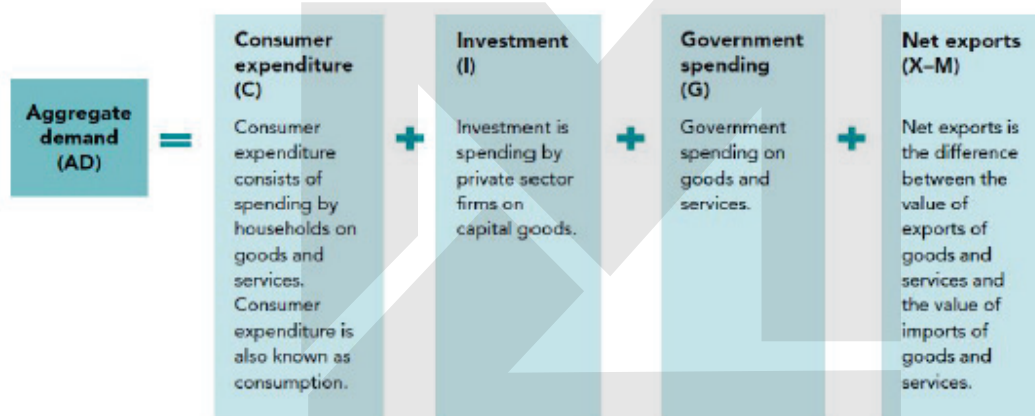


Figure 17.2: The components of aggregate demand (AD)

Determinants of the components of aggregate demand

There are a number of factors (determinants) that influence the components of demand.

Consumer expenditure

Consumer expenditure (also called consumer spending or consumption) is spending by households on goods and services to satisfy current wants, for example, spending on food, clothing, travel and entertainment.

The main influence on consumer expenditure is the level of disposable income. When income rises, total spending also usually rises, with rich people spending more than poor people.

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When a person or country is poor, most (if not all) of the disposable income has to be spent to meet current needs. Indeed, consumer expenditure may exceed income with people or countries drawing on past savings, or borrowing. This situation can be referred to as dissaving. However, when income rises, some of the income can be saved. Saving is defined as disposable income minus consumer expenditure.

A number of other factors, apart from disposable income, influence consumer expenditure. These include the distribution of income, the rate of interest, the availability of credit, expectations and wealth. If income becomes more equally distributed because of, for example, an increase in direct tax rates and state benefits, consumer expenditure is likely to rise. This is because when rich people lose income they are unlikely to cut back on their spending significantly, while people on low incomes who gain more income will spend most of the extra.

Households will also usually spend more when interest rates are low. This is because the return from saving will be reduced, buying goods on credit will be cheaper and households that have borrowed before to buy a house, for example, will have more money to spend. If it becomes easier to obtain loans, total spending is likely to increase. However, people are unlikely to borrow and to increase their spending if they are pessimistic about the future. Indeed, expectations about future economic prospects are thought to be a major influence on consumer expenditure. When people become more optimistic that their future jobs are secure and that their incomes will rise, they are likely to increase their spending. An increase in wealth, which may result, for example, from a rise in the value of houses or the price of shares, will also probably increase consumer expenditure.

Investment

Investment includes private sector spending by firms on capital goods, such as factories, offices, machinery and delivery vehicles.

The amount of private sector investment undertaken is influenced by changes in consumer demand, the rate of interest, technology, the cost of capital goods, expectations and government policy. If consumer demand rises, firms are likely to want to buy more capital equipment to expand their capacity. Similarly, a fall in the rate of interest is likely to stimulate a rise in investment. The cost of investment will fall. Firms that borrow to buy capital goods will find it cheaper and firms that use retained profits will find that the opportunity cost of investment will fall. Firms will also expect higher sales as lower interest rates will raise consumer demand.

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Advances in technology will raise productivity of capital goods and so will probably stimulate more investment. Similarly, a fall in the price of capital equipment and/or cost of installation of capital goods is likely to raise investment.

As with consumer expenditure, expectations can play a key role in determining investment. When firms are optimistic that economic conditions are improving and demand for their products will rise, they will be encouraged to raise their investment. Governments can also seek to increase private sector investment by cutting corporate tax (the tax on company profits) and by providing investment subsidies.

Government investment is usually included in the government spending component. This allows the contribution of the government sector and the difference between aggregate demand in a closed economy without a government sector and one with a government sector to be clearly seen.

Government spending

Government spending includes expenditure on providing merit goods, such as education and healthcare, and public goods, such as defence. It includes, for instance, medicines used in state hospitals, school equipment, military aircraft and government investment in infrastructure.

The amount of government spending in any period is influenced by government policy, tax revenue and demographic changes. If a government wants to raise economic activity, it may decide to raise its spending. Higher government tax revenue will enable a government to spend more, without needing to borrow. Pressure for a rise in government spending may come from an increase in the number of children (education) and/or an increase in the number of elderly people (healthcare).

Net exports

The level of net exports is influenced by the country's GDP, other countries' GDPs, the relative price and quality competitiveness of the country's products and its exchange rate. When a country's GDP rises, demand for imports usually increases. Some products may also be diverted from the export market to the domestic market. When incomes rise abroad, demand for the country's exports is likely to increase. A rise in exports may also result from an improvement in the competitiveness of the country's products, due for example to a rise in productivity or improved marketing.

The level of the exchange rate can be a key influence on net exports. If the exchange rate falls in value, the country's exports will become cheaper and imports will become more expensive. If demand for exports and imports is elastic, export revenue will rise while import expenditure will fall, causing net exports to fall.

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The aggregate demand curve

The aggregate demand (AD) curve shows the different quantities of total demand for the economy's products at different prices. A rise in the price level will cause a contraction in aggregate demand and a fall in the price level will result in an extension in aggregate demand. The downward sloping nature of the AD curve is shown in Figure 17.3.

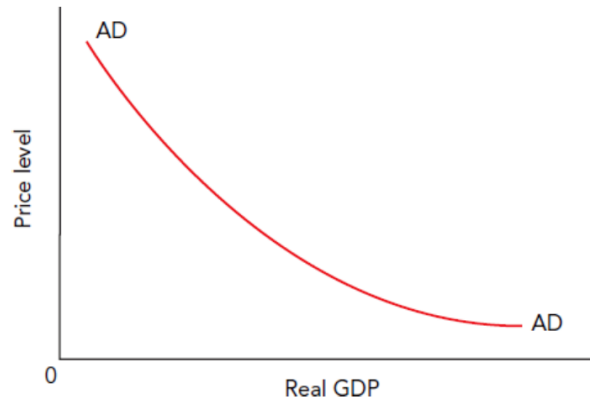


Figure 17.3: The aggregate demand curve

The relationship between aggregate demand and the price level might seem similar to the relationship shown in the demand curve for an individual product. There is, however, a significant difference. A demand curve for a product shows the relationship between a change in the relative price of a product and the quantity demanded. The price of the product is changing but it is assumed that the prices of other products have not changed. More of the product is purchased when the price falls, in part because people change from rival products. In contrast, in the case of the AD curve, the prices of most products are changing in the same direction.

So why does aggregate demand fall when the price level rises and rise when the price level falls? There are three reasons:

1. The wealth effect: A rise in the price level will reduce the amount of goods and services that people's wealth can buy. The purchasing power of savings held in the form of bank accounts and other financial assets will fall.
2. The international effect: A rise in the price level will reduce demand for net exports as exports will become less price competitive while imports will become more price competitive.
3. The interest rate effect: A rise in the price level will increase demand for money to pay the higher prices. This, in turn, will increase the interest rate. A higher interest rate usually results in a reduction in consumption and investment.

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Shifts in the aggregate demand curve

While a change in the price level causes a movement along the AD curve, if any non-price level influence causes aggregate demand to change, then the whole AD curve will shift. A shift to the left indicates a decrease in aggregate demand while a shift to the right shows an increase in aggregate demand, as shown in Figure 17.4.

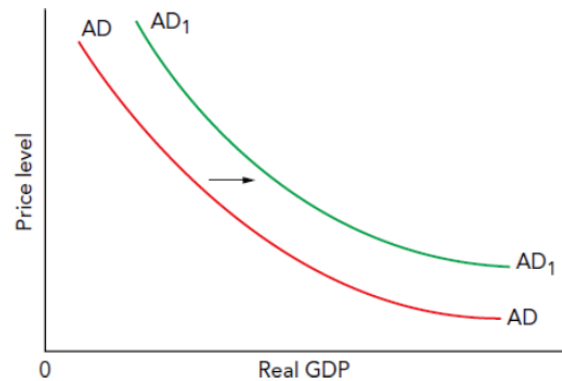


Figure 17.4: An increase in aggregate demand

A change in any non-price level influence on consumption, investment, government spending and net exports will shift the AD curve. Examples that would cause an increase in aggregate demand include:

- consumer expenditure - a rise in consumer confidence, a cut in income tax, an increase in wealth, a rise in the money supply, an increase in population
- investment - a rise in business confidence, a cut in corporate tax, advances in technology
- government spending - a desire to stimulate economic activity, a desire to win political support
- net exports - a fall in the exchange rate, a rise in the quality of domestically produced products, an increase in incomes abroad.

Aggregate supply

Aggregate supply (AS) is the total planned supply of all the producers in the country. Economists sometimes differentiate between short-run aggregate supply (SRAS) and long-run aggregate supply (LRAS). Short-run aggregate supply is the output that will be supplied in a period of time when the prices of factors of production (inputs, resources) have not had time to adjust to changes in aggregate demand and the price level.

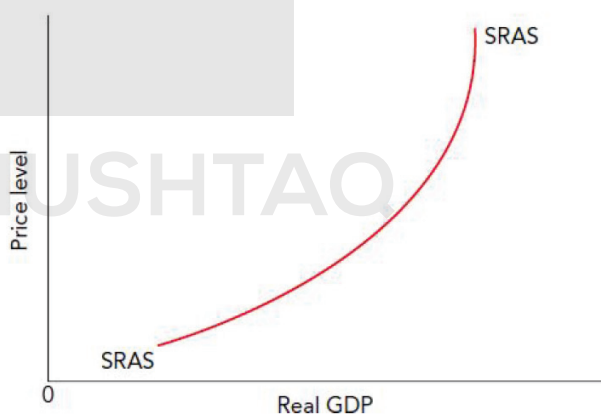


Figure 17.5: The short-run aggregate supply curve

In contrast, long-run aggregate supply is the output that will be supplied in the time period when the prices of factors of production have fully adjusted to changes in aggregate demand and the price level.

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The short-run aggregate supply curve

The short-run aggregate supply curve slopes up from left to right as shown in Figure 17.5.

As the price level rises, producers are willing and able to supply more goods and services. There are three possible reasons for this positive relationship:

1. The profit effect: As the price level (that is, the price of goods and services) increases, the prices of factors of production such as wages do not change. So as the price level rises, the gap between output and input prices widens and the amount of profit increases.
2. The cost effect: It is assumed that wage rates, raw material costs and other input prices remain unchanged along an individual SRAS curve. However, average costs may rise as output increases. This is because, for example, overtime payments may have to be paid and costs will be involved in recruiting more workers. To cover any extra costs that may be involved in producing a higher output, producers will require higher prices.
3. The misinterpretation effect: Producers may confuse changes in the price level with changes in relative prices. They may think that a rise in the price they receive for their products indicates that their own product is becoming more popular. As a result, they may be encouraged to produce more.

Shifts in the short-run aggregate supply curve

While a change in the price level will cause a movement along the short-run aggregate supply curve, there are four main causes of a shift in the SRAS curve:

1. A change in the price of factors of production: A rise in wage rates, not matched by an increase in labour productivity, and raw material costs will cause a decrease in SRAS, shifting the curve to the left as illustrated in Figure 17.6.
2. A change in taxes on firms: A reduction in corporation tax or indirect taxes will cause an increase in SRAS.
3. A change in factor productivity/quality of resources: A rise in labour productivity and/or capital productivity will cause an increase in aggregate supply both in the short and long run.

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4. A change in the quantity of resources: In the short run the supply of inputs may be influenced by supply-side shocks including natural disasters. These shocks may not have a significant impact on productive potential in the long run.

The factors that will cause an increase in the quantity of resources in the long run will also increase SRAS.

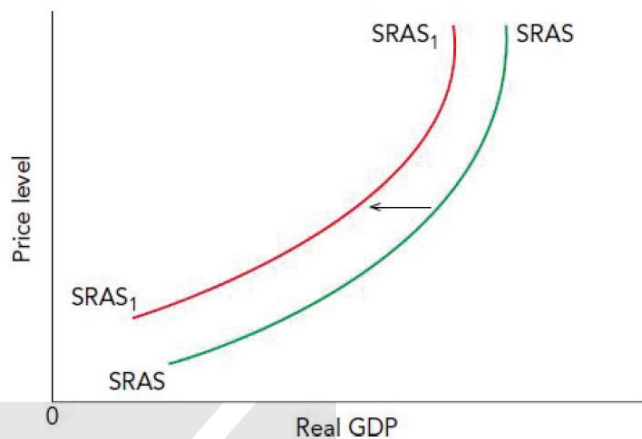


Figure 17.6: A decrease in short-run aggregate supply

The shape of the long-run aggregate supply curve

The long-run aggregate supply curve shows the relationship between real GDP and changes in the price level when there has been time for input prices to adjust to changes in aggregate demand.

A group of economists known as Keynesians often represent the LRAS curve as perfectly elastic at low rates of output, then upward sloping over a range of output and finally perfectly inelastic. This is to emphasize their view that, in the long run, an economy can operate at any level of output and not necessarily at full capacity. Figure 17.7 shows that from 0 to Y , output can be raised without increasing the price level.

When output and hence employment are low, firms can attract more resources without raising their prices. There is time for input prices to change but, due to the low level of aggregate demand, they do not. For example, when unemployment is high, the offer of a job may be sufficient to attract new workers.

As output rises from Y to Y_1 , firms begin to experience shortages of inputs and bid up wages, raw material prices and the price of capital equipment. When output reaches Y_1 , the economy is producing the maximum output it can make with existing resources.

Another group of economists, called new classical economists, illustrate the LRAS curve as a vertical line. This is because

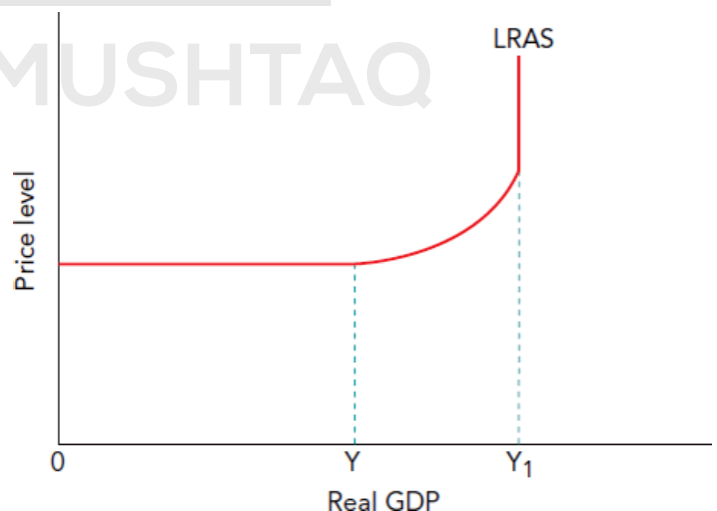


Figure 17.7: The Keynesian long-run aggregate supply curve

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they think that in the long run the economy will operate at full capacity. This version of the LRAS curve is shown in Figure 17.8.

New classical economists think that a rise in aggregate demand may increase output in the short run by encouraging firms to make more intensive use of their existing resources. Workers may be asked to work overtime and capital equipment may be operated for long periods before undertaking routine maintenance. In the long run, however, this more intensive use of resources will raise costs of production. The economy will move onto a new SRAS curve and back to the LRAS curve. Output will return to the initial level but at a higher price level, as shown in Figure 17.9.

Shifts in the LRAS curve

Both Keynesian and new classical economists agree that the causes of a shift in the LRAS curve are a change in the quantity and/or quality of resources (factor productivity). Both of these will increase the productive potential of an economy.

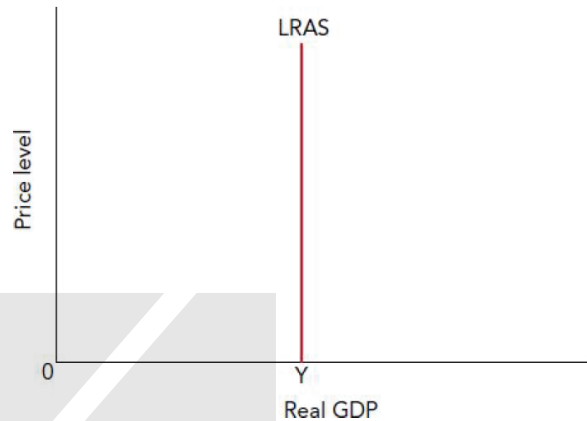


Figure 17.8: The new classical long-run aggregate supply curve

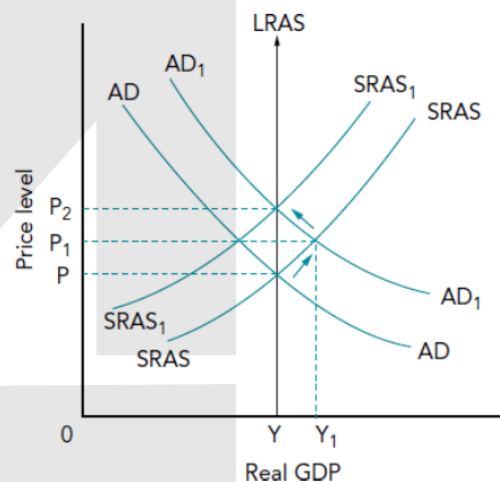


Figure 17.9: New classical view of an increase in aggregate demand

The **causes** of an increase in the quantity of resources in the long run are:

- **Net immigration:** Net immigration will increase the size of the labour force if the immigrants are of working age.
- **An increase in the retirement age:** This will increase the size of the labour force. A number of countries have raised the age at which people can receive a state pension and some of these countries plan to raise it even further in the future as life expectancy increases.
- **More women entering the labour force:** The proportion of women who work varies from country to country. For example, in 2020, only 21% of women of working age were in the labour force in Saudi Arabia, while in Norway the equivalent figure was 72%.

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- Net investment: If gross investment (total investment) exceeds depreciation (capital goods that have to be replaced because they have become worn out or out of date) there will be additions to the capital stock.
- Discovery of new resources: The discovery of, for instance, new oil fields or gold mines can increase a country's productive potential.
- Land reclamation (the creation of new land from the sea): For instance, in recent years, Dubai has added considerably to its land area by reclaiming land on which it has built apartments, hotels, marinas, theme parks and beaches.

The two main causes of an increase in the quality of resources are:

1. Improved education and training - this will increase the skills of workers and so raise labour productivity
2. Advances in technology - these both reduce costs of production and increase productive capacity.

TIP: Be careful not to confuse aggregate demand and aggregate supply diagrams with demand and supply curves. Remember to include macro-labelling on AD/AS diagrams that is price level (or general price level) rather than price and real GDP (national income and so on) rather than quantity.

Macroequilibrium and disequilibrium

The equilibrium level of output and the price level are determined where aggregate demand is equal to aggregate supply. The macroeconomic equilibrium is illustrated by the point where the AD and AS curves intersect, as shown in Figure 17.11.

If the price level was initially below P , there would be macro disequilibrium.

The excess demand would push the price level back up to the equilibrium level. If the price level was above P , some goods and services would not be sold and firms would have to cut their prices.

Changes in aggregate demand and aggregate supply will move the economy to a new macroeconomic position. Where that position will be depends on the direction of the change, the size of the change and the initial level of economic activity. For example, an economy may

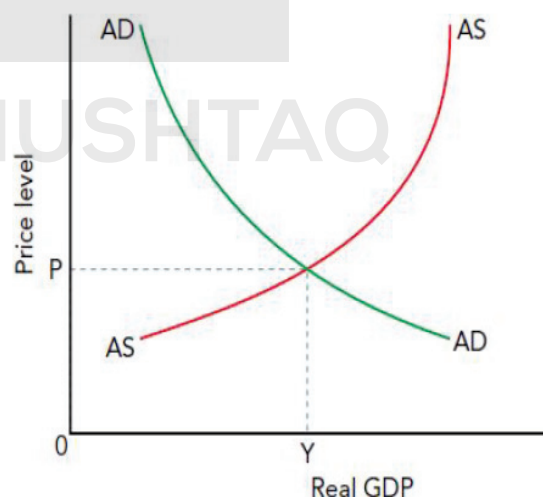


Figure 17.11: Macroeconomic equilibrium. The point of equilibrium is where AD and AS intersect

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initially be operating below but close to its productive potential. In this case, an increase in aggregate demand may increase output, employment and the price level.

Classical view	Keynesian view
LRAS is vertical at the full-employment level	LRAS is upward-sloping for a range of output below full employment
LRAS is upward-sloping for a range of output below full employment	The economy could settle at a level of output below full employment
Policy intervention is not needed because the economy adjusts rapidly	Policy intervention may be needed to move towards full employment
Policy intervention may be needed to move towards full employment	Aggregate supply is sensitive to the price level when the economy is below full employment

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